

ML Week 2

Number of participants: 171



If the loss function in regression is

1. changed to mean absolute error (MAE)

$L(y, \hat{y}) = |y - \hat{y}|$, then:

3 correct answers
out of 26 respondents

The optimal function f^* minimizing the expected prediction error is again the conditional mean.



15 votes

The model parameters fitted with the empirical MAE are the same as for MSE.



6 votes



The model is less sensitive to outliers.



12 votes



Fitting with gradient descent becomes more difficult.



9 votes



2. In linear regression

3 correct answers
out of 31 respondents



Overfitting can occur.



16 votes

We only need to keep the significant coefficients in our predictive model.



11 votes

Gradient descent cannot be used.



6 votes

RSS is a good estimation of the expected prediction error.



15 votes



3. In gradient descent for optimization of loss functions:

8 correct answers
out of 29 respondents

It is always better to choose a small learning rate.



6 votes



A large learning rate can make the optimization fail.



21 votes



The initial value can have an influence on the optimal value obtained by gradient descent.



17 votes

Gradient descent should not be used in linear models since there we have closed-form solutions.



2 votes



4. Which of the following is true about linear regression?

19 correct answers
out of 31 respondents



It can be applied to problems where the response depends non-linearly on the predictors.



21 votes

The regression coefficients can only be estimated with gradient-descent.



2 votes

Without data standardization, it is impossible to fit a linear regression model.



10 votes